SCIENCE & TECHNOLOGY PART 2 | CLASS 10 | MAHARASHTRA SSC BOARD

Chapter 10: Disaster Management

MODEL ANSWER SHEET / ANSWER KEY

29	2	4
QUESTIONS	IN TEXTBOOK	DIAGRAMS

HOW TO USE
Cover the answer box ✓ and answer first. A **TEXTBOOK** *strip means the same question is in your Balbharati Science-2 textbook, at the set and page shown. Tick ☐ each section once revised.*

01	02	03	04
Name It	Define	One Sentence	Solve
1Q	1Q	2Q	25Q

01 Name the following

1 QUESTIONS ☐ Revised

Q1 Name the type of disaster.

Disaster	Type
Tsunami, volcanic eruption, earthquake	_____
Cyclone, snow storms, droughts, floods	_____
Forest fire, weed, fungal disease spreading	_____
Communicable virus, bacteria, bite of poisonous animal	_____
Poisonous gases, atomic test, unplanned action, accident	_____

✓ ANSWER / SOLUTION

Disaster	Type
Tsunami, volcanic eruption, earthquake	**Geophysical: Geological**
Cyclone, snow storms, droughts, floods	**Geophysical: Atmospheric**
Forest fire, weed, fungal disease spreading	**Biological: Plant**
Communicable virus, bacteria, bite of poisonous animal	**Biological: Animal**
Poisonous gases, atomic test, unplanned action, accident	**Man-made-Unknown**

02 Define

1 QUESTIONS ☐ Revised

Q1 First aid

ANSWER / SOLUTION

The primary help offered to the victims of disasters before actual medical treatment is available is known as first aid.

03 Answer in one sentence

2 QUESTIONS ☐ Revised

Q1 Complete the following table.

Type of Disaster	Effects
A. _____	Injuries, emotional and mental stress, Outbreaks of epidemics, deaths of victims.
B. Environmental	_____

ANSWER / SOLUTION

Type of Disaster	Effects
A. **Medical**	Injuries, emotional and mental stress, Outbreaks of epidemics, deaths of victims.
B. Environmental	**Partial or complete collapsing of buildings, drying out of water resources, contamination of water. Pollution of environment due to decomposing corpses of humans and other animals.**

Q2 Correct the given statement and rewrite it.

An effect of disaster on economic leadership is that if local leadership is not strong enough, citizens get confused.

ANSWER / SOLUTION

An effect of disaster on social leadership is that if local leadership is not strong enough, citizens get confused.

04 Solve the following questions

25 QUESTIONS · 2 ☐ Revised

Q1 State three aspects of disaster which are important in view of common citizens.

ANSWER / SOLUTION

The three aspects of disaster which are important in view of common citizens are as follows:

i. Phase of emergency
ii. Transitional phase
iii. Reconstruction phase

Q2 Write the effects of political disaster.

ANSWER / SOLUTION

The effects of a political disaster are as political unrest, loss of – political trust, political legitimacy and political stability, etc.

Q3 Define disaster.

ANSWER / SOLUTION

The unpredictable dangerous events which occur many times in the environment are known as disasters.

Q4 Write a short note on Phase of emergency.

ANSWER / SOLUTION

Phase of emergency:

i. Maximum lives can be saved by quick action during this phase.
ii. Search and rescue operations, medical assistance, first aid, restoring communication services, removing the people from the affected area, etc. are actions expected in this phase.
iii. The actual gravity of the disaster can be estimated in this phase only.

Q5 What are the seven main aspects of disaster management cycle?

ANSWER / SOLUTION

The seven main aspects of disaster management cycle are;

i. Preparation
ii. Redemption
iii. Preparedness
iv. Impact of disaster
v. Response
vi. Resurgence
vii. Restoration

Q6 Give two examples of Mock Drill.

ANSWER / SOLUTION

i. Fire mock drills are arranged in various schools by fire fighters, which includes demonstrations like extinguishing the fire, rescuing the people trapped at higher floors of buildings, rescuing the persons whose clothing have caught fire, etc.
ii. Earthquake mock drills are arranged which includes timely evacuation (if possible), providing information on where to seek shelter, how to protect heads and bodies from falling objects, etc.

Q7 Write four objectives of Mock Drill.

ANSWER / SOLUTION

The different objectives of mock drill are as follows:

i. To evaluate the response to the disaster.
ii. To improve the coordination between various departments of disaster control.
iii. To identify the abilities of the organization and its people for facing disaster.
iv. To check the competency of the planned actions.
v. To identify the possible errors and risks.

Q8 How would we rescue the citizens whose clothes had caught fire?

ANSWER / SOLUTION

We can rescue the citizens whose clothes have caught fire in the following manner:

i. Stop the person from running and advise them to drop and roll on the floor to put out the fire
ii. Dousing the fire with water
iii. Using a fire blanket, non-flammable mat, blanket or article of clothing to assist in smothering the fire.
iv. Treat the burn injury appropriately once the fire is out and wait for help.

Q9 Enlist the necessary materials in the First-Aid kit.

ANSWER / SOLUTION

i. A first aid kit is a small box that contains items required for administering first aid to a victim of an accident or disaster.
ii. Some of the important materials that should be included in a first aid kit are:
    i. Bandage strips of different sizes, triangular and circular bandage,
    ii. wound gauze,
    iii. medicated cotton,
    iv. hand gloves,
    v. clean and dry cloth pieces,
    vi. soap,
    vii. antiseptic (Dettol/Savlon),
    viii. safety pins,
    ix. blade,
    x. small pins,
    xi. needle,
    xii. torch,
    xiii. scissor,
    xiv. thermometer,
    xv. petroleum jelly, etc.

Q10 Give an example of disaster due to unawareness.

ANSWER / SOLUTION

Fires, explosions, electric shocks, burns, heat shock, fire due to electric short circuit, etc. are few examples of disasters caused due to unawareness.

Q11 Give two examples of Post-disaster management.

ANSWER / SOLUTION

Post-disaster management includes the following:

i. Providing help to the victims of disasters, in all possible ways.
ii. Arranging help to the victims of the disaster with the help of local people who were not harmed by the disaster.
iii. Quick establishment of control and help centres depending upon the type of disaster.
iv. Categorization and delivery of relief materials obtained from control centres and continuous review of the same.
v. Being alert and prepared for disaster rescue.

Q12 Give one example each of a huge disaster and small disaster.

ANSWER / SOLUTION

i. **Huge disaster:** Landslide, earthquake, volcano, tsunami, flooding, forest fire, snow storms, cyclones, droughts, meteorite, war, etc.
ii. **Small disaster:** Electric shock, minor burns, heat shock, etc.

Q13 Give one example each of long duration disaster and short duration disaster.

ANSWER / SOLUTION

i. **Long duration disaster:** Landslide, earthquake, volcano, tsunami, Flooding, forest fire, snowstorm, cyclones, droughts, meteorite, war, etc.
ii. **Short duration disaster:** Electric shock, heat shock, minor burns, etc.

Q14 In account of disaster, which important facts are to be considered?

ANSWER / SOLUTION

In account of disaster, the important facts to be considered are as follows:

i. Pre-disaster phase
ii. Warning phase
iii. Emergency phase
iv. Rehabilitation phase
v. Recovery phase
vi. Reconstruction phase

Q15 What is the transitional phase of disaster?

ANSWER / SOLUTION

The features of the transitional phase of disaster management are as follows:

i. Rehabilitation work is started in this phase.
ii. It includes clearing of debris, restoring water supply, repairing roads, etc. in order to bring normalcy in public life.
iii. Rehabilitation of the victims is an important aspect of this phase. Different institutes offer monetary and other types of help to the victims.
iv. In order to soothe the mental stress of the victims at the earliest, they are offered with permanent means of earning livelihood also known as true rehabilitation.

Q16 Which disastrous effects occur during earthquakes?

ANSWER / SOLUTION

Some of the disastrous effects of the earthquake are:

i. Collapsing of houses, developing cracks in land, etc.
ii. Earthquake leads to massive destruction of life and property.
iii. It leads to the destruction of infrastructure viz. roads, bridges, etc.
iv. Rivers tend to change the course of flow due to the blocking of the natural course of water.

Q17 What is a catastrophic disaster? Give an example.

ANSWER / SOLUTION

Disasters that lead to total chaos and huge loss of life and property in respective region are known as catastrophic disaster.
e.g. Cyclones in Odisha, catastrophic earthquakes of Gujarat and Latur, frequently buzzing cyclones in coastal Andhra Pradesh, etc.

Q18 Mention the long term effects of disaster on society.

ANSWER / SOLUTION

long term effects of disaster on society:

1. **Social Effects:** Loss of life, displacement, disruption of education and healthcare, and weakened social order.
2. **Economic Effects:** Infrastructure destruction, unemployment, inflation, and increased government expenditure.
3. **Environmental Effects:** Loss of biodiversity, pollution, deforestation, and climate change impact.
4. **Psychological Effects:** Trauma, PTSD, anxiety, and community disintegration.

Disasters deeply impact society, requiring long-term recovery and resilience efforts.

Q19 Which are very dangerous environmental events?

ANSWER / SOLUTION

Very dangerous environmental events are disasters like floods, wet and dry famine, cyclones, earthquakes, volcanoes, etc.

Q20 Which are natural disasters?

ANSWER / SOLUTION

The main natural disasters are floods, wet and dry famine, cyclones, earthquakes, volcanoes, etc.

Q21 Explain the symbol.



TEXTBOOK **Exercise** · Q.8 (symbol 2) · **Page 120**

ANSWER / SOLUTION

Explanation of the Symbol: This symbol represents highly toxic or poisonous substances, which are extremely dangerous even on a single exposure. Their action on the human body is very rapid and can be fatal (lead to death).

Disaster that may occur: If this symbol is ignored, it can lead to a serious chemical disaster, a hazardous gas leak accident, or an intentional health disaster.

Q22 Explain the symbol.



TEXTBOOK **Exercise** · Q.8 (symbol 3) · **Page 120**

ANSWER / SOLUTION

Explanation of the Symbol: This symbol represents environmental hazards, indicating substances that are highly toxic to aquatic life (fish) and nature (trees) with long-lasting harmful effects.

Disaster that may occur: If this symbol is ignored, it can lead to an environmental disaster or ecological collapse, causing severe pollution of water bodies, soil contamination, and the mass death of aquatic organisms.

Q22 In 2020, the whole world is facing a disaster.

a. Which type of disaster it is?
b. Write the name of the virus causing this disaster.
c. Which four rules you should follow to survive from this disaster?
d. Write any four effects of this disaster you observed.

ANSWER / SOLUTION

a. It is a natural (biological) disaster
b. Covid – 19 (Coronavirus)
c. **Rules to follow in order to survive this disaster:**
   i. Use masks
   ii. Wash hands and face with soap and water after returning home from outside.
   iii. Avoid visiting overcrowded places
   iv. Use hand sanitizers regularly
   v. Avoid shaking hands with others
   vi. Maintain social distancing in public spaces.
d.
   i. Alarmingly high number of deaths due to the virus
   ii. Economic losses
   iii. Unemployment
   iv. Closure of educational institutes resulting in academic losses
   v. Social, financial, political, and mental unrest

Q23 **Observe the given photographs and answer the questions.**



a. Name the disaster shown in the pictures.
b. Write any two effects of this disaster.

ANSWER / SOLUTION

a. The disaster shown in the pictures is floods.
b. The effects of floods are the collapsing of bridges, flooding of coastal villages, massive loss of life and property, shortage of food, soil erosion, spread of water-borne diseases and other epidemics.

Q24 **Observe the given pictures and answer the questions.**



a. Identify the disasters shown in both pictures.
b. Write their effects.
c. Which primary precautions will you take in both situations?

ANSWER / SOLUTION

a. Earthquake is shown in image 'A' and fire is shown in image 'B'.

b. The effects of these disasters are loss of life, financial crisis, damage to property and mental stress.

c.

In case of earthquake:

i. Run for a safe exit and try to reach an open place away from buildings and tall trees.
ii. Avoid using elevators while exiting a building. Use staircase instead.
iii. If you are stuck in a room, take shelter under a hard surface, such as a table.

In case of fire:

i. Use a fire extinguisher to put off the fire.
ii. Run towards a safe place away from the location of fire.
iii. While entering to the safe room, make sure that there is no fire in that room. You can do so by touching the door and feeling the heat on door before opening it.
iv. If there is lot of smoke, keep your head low to the ground to avoid excess inhalation of smoke.
v. Call the fire brigade immediately.
vi. **Also, if any person has suffered minor burns,**
a. Wash the injured part with water or hold it under water.
b. Clean the wound with an antiseptic solution using a cotton swab and apply ointments that are specially prescribed for burns.
c. Cover the wound using dry dressing.
If a person has suffered major burns,
a. Use sterilized cloth to cover the wound.
b. Immediately remove jewellery, shoes, etc., from the body.
c. Do not touch or burst the blisters on the skin, remove the cloth stuck to the region of burnt skin.